

IK 523: Ancient Indian Mathematics

Course Code: IK 523

Course Name: Ancient Indian Mathematics

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: B.Tech/MTech/MS/MSc/MA/PhD

Mutual Exclusion: None

Approval: 64th BoA

Preamble

The course introduces the foundational mathematical traditions of ancient India including geometry in Śulbasūtras, decimal place value systems, numeral systems, arithmetic, algebra, trigonometry, cyclic quadrilaterals, Kuṭṭaka methods, continued fractions, and the Kerala School of mathematics.

The course systematically examines the contributions of major Indian mathematicians including Āryabhaṭa, Brahmagupta, Bhāskara, Mādhava, Nīlakaṇṭha, and the Kerala School traditions. Particular emphasis is placed on mathematical reasoning, computational techniques, geometrical constructions, infinite series, trigonometric developments, and the interdisciplinary significance of Indian mathematical traditions within Indian Knowledge Systems.

The course is research-oriented and emphasizes analytical learning through mathematical demonstrations, textual engagement, historical interpretation, and computational understanding of traditional Indian mathematical methods.

Course Modules with Quantitative Lecture Hours

- **Module 1: Geometry in Śulbasūtras and Foundations of Indian Mathematics:** Geometry in Śulbasūtras, altar geometry and ritual constructions, expressions for surds in Śulbasūtra texts, methods for obtaining perpendicular bisectors, cord-folding techniques, Bodhāyana method of square construction, Bodhāyana–Pythagorean theorem, and foundations of geometrical reasoning in ancient Indian mathematics. (6 Hours)
- **Module 2: Decimal Place Value System and Indian Numeration Traditions:** Numbers in Vedic literature, enumeration of large numbers from koti to mahaugha, akṣauhiṇī and named numeral systems, decimal place value system, Āryabhaṭan numeration, Bhūtasāṅkhyā system, Kaṭapayādi system, and the evolution of Indian numeral notation traditions. (6 Hours)
- **Module 3: Arithmetic, Algebra and Trigonometry in Classical Indian Mathematics:** Methods for square roots and cube roots, Āryabhaṭa's sine-table, trigonometric modifications in Tantrasaṅgraha, Mādhava series for sine and cosine functions, methods from Karaṇapaddhati for obtaining sine values, Brahmagupta's treatment of positive, negative, and zero quantities, linear and quadratic equations, cyclic quadrilaterals, and the development of algebraic and trigonometric traditions in India. (6 Hours)
- **Module 4: Līlāvati Traditions and Mathematical Applications:** Līlāvati traditions of mathematical pedagogy, arithmetical operations, inversion methods, rule of supposition, quadratic equations, mixtures, combinations, progressions, plane figures, applications of right triangles, Sūc problems, construction of quadrilaterals, cyclic quadrilaterals, value of π , area of circles, surface area and volume of spheres, and practical mathematical applications in classical India. (6 Hours)

- **Module 5: Kuṭṭaka Methods, Continued Fractions and Computational Traditions:** Kuṭṭaka methods of Āryabhaṭa and Brahmagupta, computational procedures in Indian mathematics, Vallyupasaṃhāra and continued fractions, Vallyupasaṃhāra Method I and II from Karaṇapaddhati, nearest-integer continued fractions in Dṛkkaraṇa traditions, and algorithmic traditions in classical Indian computation systems. (6 Hours)
- **Module 6: Kerala School of Mathematics and Infinite Series Traditions:** Kerala School contributions to mathematics, Mādhava series for π , Antyaśaṃskāra (end-correction), fast convergent series for π , Putumana-Somayājī series, Nīlakaṇṭha's treatment of the irrationality of π , infinite geometric series traditions, and the global significance of Kerala mathematical innovations. (6 Hours)
- **Module 7: Comparative, Computational and Interdisciplinary Perspectives on Indian Mathematics:** Comparative study of Indian and global mathematical traditions, mathematical logic and algorithmic thinking in Indian knowledge systems, computational methods in classical Indian mathematics, intersections between mathematics and astronomy, relevance of Indian mathematics to modern computational sciences and AI, and contemporary interdisciplinary applications and research perspectives. (6 Hours)

Text Books

1. Datta, B. and Singh, A. N. *History of Hindu Mathematics*, Vol. I & II, Asia Publishing House.
2. Joseph, George Gheverghese. *The Crest of the Peacock: Non-European Roots of Mathematics*, Princeton University Press.

References

1. Āryabhaṭa. *Āryabhaṭīya*.
2. Brahmagupta. *Brāhmasphuṭasiddhānta*.
3. Bhāskarācārya. *Līlāvati*.
4. Tantrasaṅgraha and Karaṇapaddhati traditions.
5. Studies on Kerala School mathematical traditions.
6. Sarasvati Amma, T. A. *Geometry in Ancient and Medieval India*, Motilal Banarsidass.
7. Śarma, K. V. Sarma. *Indian Astronomy and Mathematics*, Motilal Banarsidass